

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) specializing in distributed systems, backend infrastructure, and production observability. Built event-driven systems handling **2,000 concurrent requests/second** with zero message loss using Kafka and Temporal.io state machines. Shipped **Grafana and Sentry monitoring** across production deployments at Isomer AI; reduced pipeline latency 20% with Kafka + Spark at Capgemini. Comfortable owning reliability end-to-end -- from architecture through to on-call. Currently ramping on Go; strong in Python, TypeScript, and C.

CORE COMPETENCIES

Distributed Systems

Kafka & Event-Driven Architecture

Observability & Grafana

Backend Infrastructure

Kubernetes & Cloud (AWS / GCP)

Python & Go (learning)

Reliability & Fault Tolerance

Production Software Delivery

WORK EXPERIENCE

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Engineered distributed data pipelines using **Hadoop + Spark** for batch processing and integrated **Kafka streaming** for real-time data ingestion -- reduced ticket-processing latency by **20%** and end-to-end processing time by 15%
- Led end-to-end SDLC for automated support ticket analysis system; designed pipeline architecture, implemented fault-tolerant ingestion, and validated output quality in production

Isomer AI May 2025 - Present

Product Engineer Intern

- Built and operated production backend systems including **Temporal.io workflow orchestration**, real-time event pipelines, RBAC, document processing -- **394 commits, 7 production deployments**; integrated **Sentry observability** for production health monitoring
- Shipped **Grafana**-integrated monitoring across AWS/GCP infrastructure; designed retry logic, circuit patterns, and alerting for distributed service reliability

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Automated deployments on **AWS and GCP**; integrated **Grafana monitoring dashboards** for production workloads; supported model validation and red-teaming for reliability on enterprise insurance systems

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed and deployed 500GB cloud data platform on **Google Cloud Run** (100+ researchers worldwide); applied time-series analysis and hyperparameter tuning to improve model accuracy 12%

PROJECTS

AlgoPulse -- Distributed Event-Driven Platform Independent, 2025

Identified API-polling bottleneck under concurrent load and re-architected to event-driven streaming. Used **Temporal.io state machines** for durable workflow orchestration and **Kafka** for fault-tolerant message brokering -- sustained **2,000 async requests/second** with zero message loss and full fault recovery under failure injection.

TypeScript · Python · Temporal.io · Kafka · Supabase Edge Functions · Node.js

LLM Eval Harness -- Distributed Benchmark Pipeline Independent, 2025

Built async multi-provider evaluation pipeline with configurable concurrency, per-provider retry middleware (full-jitter exponential backoff on 429/5xx), JSONL raw logging, and **Parquet-normalized analytics**. DuckDB + Streamlit dashboard tracks p50/p95 latency, cost/call, and reliability metrics per run.

Python · DuckDB · Parquet · Streamlit · AsyncIO · Docker

Demand Paging -- Xinu OS Kernel Systems Programming, 2024

Implemented 8-store virtual memory system in **C** for **Xinu OS** with Second-Chance page replacement; reduced page faults 25%. Added 7+ kernel-level system calls with inverted page tables and custom fault handler, improving memory throughput 15%.

C · Xinu OS · Virtual Memory · Kernel Systems

Glance AI -- Privacy-First RAG System Independent, 2025

Designed per-tenant data isolation at the vector retrieval layer with deterministic output architecture -- compliance-ready for fintech/healthcare environments where data isolation is a hard requirement.

Python · Vector Databases · RAG · LangChain · LLMs

EDUCATION

MS Computer Science NC State University	Aug 2024 - May 2026
Operating Systems, Algorithms, Data Science, Distributed Systems, Software Engineering	
B.Tech Computer Science Mumbai University	Aug 2020 - May 2024
CGPA 9.32/10 · Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain	

SKILLS

Languages: Python, TypeScript, JavaScript, C, Java, SQL, R · Go (learning)

Distributed Systems: Kafka, Temporal.io, Spark, Hadoop, Redis, Kubernetes, Docker, CI/CD

Cloud & Observability: AWS, GCP, Grafana, Sentry, Prometheus (concepts), Cloud Run, Terraform

Backend: Node.js, PostgreSQL, Prisma, Express, REST, Distributed Systems, Microservices

Awards: Top 10 in 6+ national hackathons · Google Developer Student Club Lead (500+ students) · IEEE Published